Related Works					
Category	**Paper 1**	**Paper 2**	**Paper 3**	**Paper 4**	**Paper 5**
Title	Beyond Pixel: Superpixel-Based MRI Segmentation Using Machine Learning and GCN	Context-Aware Graph Neural Network for Skin Lesion Classification	Graph Attention Networks for Skin Lesion Classification with CNN-Based Node Features	SPX-GNN: An Explainable GNN for Long-Range Dependencies in Tuberculosis Classification	Semi-Supervised GNN for Multi-Modal Skin Lesion Classification
Authors	Zakia Khatun et al.	Muhammad Azeem et al.	Ghadah Naif Alwakid et al.	Muhammed Ali Pala & Muhammet Burhan Navdar	Thi Trang Nguyen et al.
Year	2024	2026	2025	2025	2026
Dataset	Custom ankle MRI (76 subjects)	ISIC2024, HAM10000, PAD-UFES-20, HIBA	DermaMNIST-Corrected	Chest X-ray (4,200 samples)	ISIC 2020
Application	Achilles tendon segmentation	Skin lesion classification	Multi-class skin lesion classification	Tuberculosis classification	Skin lesion classification
Method / Model	SLIC superpixels + RF, SVM, and 2-layer GCN	CNN + patient information + GNN with APPNP	SLIC superpixels + EfficientNet-B0 + 5-layer GAT	SLIC superpixels + LBP/Hu features + 4-layer GCN	k-NN graph + GINE + trust-based pseudo-labeling
Graph Construction Method	Images are segmented into superpixels using the SLIC algorithm. Each superpixel is a node, and edges connect neighboring superpixels (unweighted).	CNN features and metadata are concatenated and partitioned into contiguous subspaces, each acting as a node. These nodes form a fully connected graph.	Images are over-segmented using SLIC into superpixels. These form nodes of a Region-Adjacency Graph (RAG), with edges created between touching regions.	Images are decomposed into SLIC superpixel nodes. Edges are established based on spatial adjacency (shared boundaries) between regions.	Each lesion sample is a node. Edges are constructed using a k-nearest neighbor (k-NN) strategy based on the cosine similarity of multimodal features.

GCN type	A two-layer Graph Convolutional Network (GCN) architecture.	A GNN using inductive neighborhood aggregation (mean/max) and the APPNP algorithm for long-range dependencies	A five-layer multi-head Graph Attention Network (GAT).	A four-layer Graph Convolutional Network (GCN) architecture	Evaluated GCN, GAT, GraphSAGE, and GraphSAGE–GAT; the GINE backbone was selected as the final model.
Metrics	AUC, Sensitivity, Specificity, Balanced Accuracy	Accuracy, Precision, Recall, F1-score	Accuracy, AUC, Precision, Recall, F1-score	Accuracy, F1, ROC-AUC, MCC, Precision, Recall, Specificity, Zero-to-One Loss	Sensitivity, Specificity, Macro F1, Macro AUC
Strengths	Works well with noisy images and needs less data.	Combines image features with patient information.	GAT focuses more on important image regions.	High performance and provides visual explanations.	Uses unlabeled data and checks prediction reliability.
Limitations	Small dataset may limit GCN performance.	Too many graph nodes may cause overfitting.	Depends on SLIC quality and affected by class imbalance.	Class imbalance and background removal are challenging.	Missing patient information and uncertainty estimation may reduce performance.
Research Gap	Needs more testing on larger datasets and other medical images.	Relationships between different feature types need better modeling.	More effective graph construction methods need to be explored.	More work is needed to model long-range relationships and handle imbalance.	More reliable multimodal and uncertainty-aware methods are needed.
Relation to my Work	Shows how images can be converted into graphs using SLIC.	Shows how visual and patient information can be represented as graph nodes.	Closely related to my work because it uses superpixels and GAT for skin lesions.	Shows the use of superpixel graphs and GCN for medical images.	Provides a different sample-level graph approach that can be compared with image-level graphs.

Research Gap/ Problem statement:

Previous studies have used different methods to create graphs from medical images. Some methods connect nearby regions based on their spatial location, while others connect regions based on their visual or feature similarity. However, there are limited studies that compare these different graph construction methods on the same skin lesion dataset. Therefore, it is still unclear which type of graph structure is more effective for skin lesion classification. This study addresses this gap by using the HAM10000 dataset to compare spatial-based, visual-similarity-based, and hybrid graph construction methods and determine which approach works best for GNN-based skin lesion classification.